Quiz # 2 (Data Structures) CLO2 **BS DS Fall 2025 Morning**

Name:_____ **Roll number:_____**

Time Allowed: 20 min **Total Marks: 30**

Q1. Write asymptotically equal algorithms in a single line: [10 Marks]

$3n^3 - 25n + 16$, $\frac{5n^2+500\,n+16}{2n}$, $100n^2lg_3n + 2500$, $13n^2 - 100006$, $n^3 + n^2lgn$, 2n-250, $\sqrt[2]{n^{\frac{2}{3}}} + 25$

1. $3n^3 - 25n + 16$, $n^3 + n^2lgn$
2. $\frac{5n^2+500\,n+16}{2n}$, 2n-250
3. $100n^2lg_3n + 2500$
4. $13n^2 - 100006$
5. $\sqrt[2]{n^{\frac{2}{3}}} + 25$

Q2. Find time equation and asymptotic running time: [10+10= 20 Marks]

```
for (i = 1; i <= 2000 ; i++)
   for (j = 1; j <= n ; j = j * 4)
      a = i + j;
```

2001 (2000 times true and 1 time false)
2000 ($\log_4$ n + 1)
2000 ($\log_4$ n)

T(n) = 2001 +2000 **$\log_4$ n** + 2000 + 2000 **$\log_4$ n**

T(n) = 4000 **$\log_4$ n** + 4001

Asymptotic Time: 4000 $\log_4$ n = O($\log_4$ n)

```
i=1
while (i<=n)
    j=n
    while (j>=i)
          a=i+j
          j - -
    i + +
```

n+1 (n times true and 1 time false)
n
$\sum_{i=1}^{n} n - i + 2$
$\sum_{i=1}^{n} n - i + 1$
$\sum_{i=1}^{n} n - i + 1$
n

T(n) = n + 1 + $\sum_{i=1}^{n} n - i + 2$ + $\sum_{i=1}^{n} n - i + 1$ + $\sum_{i=1}^{n} n - i + 1$ + n

T(n) = n + 1 + $\sum_{i=1}^{n} n - i + 2$ + $\sum_{i=1}^{n} n - i + 1$

T(n) = n + 1 + $\frac{(n+1)(n+2)}{2} - 1$ + $\frac{(n)(n+1)}{2}$ + $\frac{(n)(n+1)}{2}$ + n

T(n) = An^2 + Bn+ C